CHAPTER 2

Principles of Layering

Why we layer · service, interface, protocol · encapsulation · port vs IP vs MAC · which device reads which header

QUICK TRICK

Going **down** the stack: each layer **adds a header** (Link also adds a trailer). Going **up**: each layer **removes** its header. Port = which *process*. IP = which *host* (end to end). MAC = which *next hop* (changes every link).

2.1 Why layers?

● **LEVEL 1 · BEGINNER**

The problem. Sending data across the world involves cables, errors, addresses, routes, reliability and applications. One giant program would be impossible to change. **The idea.** Split the job into layers. Each layer solves one problem and uses only the layer just below it.

Analogy. Posting a letter. You write the letter (application). The post office puts it in a bag with a city code (network). The truck driver only looks at the road (link). You do not care how the truck works, and the driver does not read your letter.

2.2 Service, interface, protocol

● **LEVEL 2 · EXAM**

Term	Meaning	Direction
Service	What a layer offers to the layer above (e.g. reliable byte stream).	Vertical, same machine
Interface	How the upper layer asks for that service (e.g. socket calls).	Vertical, same machine
Protocol	Rules that two **peer** layers on different machines follow (e.g. TCP between two hosts).	Horizontal, across machines

QUICK TRICK

Memory line: **Service = up and down. Protocol = side to side.** A layer can change its protocol without changing the service it offers.

2.3 Encapsulation

● **LEVEL 2 · EXAM**

```
Sender                                          Receiver
Application   [ DATA ]                  ------->  [ DATA ]
Transport     [TCP hdr][ DATA ]          ------->  segment
Network       [IP hdr][TCP hdr][ DATA ]  ------->  packet / datagram
Link          [Eth hdr][IP][TCP][DATA][Eth trailer]  frame
Physical      0101010101...  (bits on the wire)
```

Layer	PDU name	Address used	Typical size of its header
Application	Message / data	Name (URL, domain)	varies
Transport	Segment (TCP) / datagram (UDP)	Port (16 bit)	TCP 20 B+, UDP 8 B
Network	Packet / datagram	IP (32 bit)	IPv4 20 B+
Link	Frame	MAC (48 bit)	Ethernet 14 B + 4 B trailer (FCS)
Physical	Bits	none	none

Overhead and efficiency. Efficiency = useful data ÷ total bits sent. Example: 1000 B data + TCP 20 + IP 20 + Ethernet 18 = 1058 B, so efficiency = 1000/1058 ≈ **94.5%**.

2.4 Which address works where?

● LEVEL 2 · EXAM

Address	Identifies	Scope	Changes along the path?
Port	A process / application	End to end (host to host process)	No (except at a NAT)
IP	A host interface	End to end	No (except at a NAT)
MAC	A NIC on one link	Hop to hop (one LAN)	**Yes, at every router**

```
Host A --- LAN1 --- Router R --- LAN2 --- Host B
Frame on LAN1 : src MAC = A      dst MAC = R(left)    src IP = A  dst IP = B
Frame on LAN2 : src MAC = R(rt)  dst MAC = B          src IP = A  dst IP = B
                ^ MAC changes at each hop             ^ IP stays the same
```

2.5 Devices and the header they read

● LEVEL 2 · EXAM

Device	Layer	Reads	Collision domains	Broadcast domains
Repeater / Hub	1	nothing (bits)	one for all ports	one
Bridge / Switch	2	MAC address	one per port	one
Router	3	IP address	one per port	one per interface

2.6 Common mistakes

COMMON MISTAKE

- Saying the IP address changes at each router (it is the **MAC** that changes).
- Mixing service and protocol (service is vertical, protocol is horizontal).
- Forgetting the Ethernet trailer (4 B FCS) when computing overhead.
- Believing a router reads TCP ports (a normal router does not; a NAT does).

QUICK TRICK

Memory line: "Port = process, IP = host, MAC = next hop."

GATE QUICK CHECK — Chapter 2

```
PDU      : app = data | transport = segment | network = packet | link = frame | phy = bits
Header   : TCP 20 B | UDP 8 B | IPv4 20 B | Ethernet 14 B + 4 B FCS
Address  : port 16 b | IP 32 b | MAC 48 b   (MAC rewritten at every router)
Device   : hub L1 | switch L2 | router L3
```

Practice questions

● LEVEL 1 · BEGINNER

Q1. What is the PDU at the transport layer for TCP called?

Q2. Which address changes at every router: port, IP or MAC?

Q3. Which address identifies the receiving *process*?

Q4. Which layer provides process-to-process delivery?

● LEVEL 2 · EXAM

Q5. A packet goes from host A to host B through one router (no NAT). Which of source IP, destination IP, source MAC, destination MAC change?

Q6. Efficiency when 512 B of data is sent with TCP 20 B, IP 20 B and Ethernet 18 B overhead.

Q7. A frame crosses 3 routers between two hosts (all links Ethernet). How many different Ethernet frames carry this packet?

Q8. Difference between a service and a protocol.

● LEVEL 3 · GATE

Q9. How many collision domains and broadcast domains exist if 6 hosts connect to a single switch, and the switch connects to a router with one other interface leading to a hub with 4 hosts?

Q10. Which header does a layer-2 switch examine when forwarding?

Answers

Q	Solution
1	A **segment**.
2	**MAC** (source and destination MAC are rewritten on each link).
3	The **port number**.
4	The **transport** layer.
5	Only **source MAC and destination MAC** change. Source and destination IP stay the same.
6	512/(512+58) = 512/570 ≈ **89.8%**.
7	**4**: one per link (source host, then each of the 3 routers builds a new frame).
8	A **service** is what a layer offers the layer above (vertical, same machine). A **protocol** is the rule set peers on different machines follow (horizontal).
9	Collision domains: 6 (switch ports to hosts) + 1 (switch to router) + 1 (router to hub, which includes the 4 hosts) = **8**. Broadcast domains: **2** (one per router interface).
10	The **Ethernet (MAC) header**: destination MAC address.

Next: Chapter 3 — Switching and Performance Metrics (circuit, packet, virtual circuit; delay arithmetic).

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